PRODUCING A TOTALLY RAMIFIED FINITE GALOIS EXTENSION THAT EITHER SOLVES THE RAMIFICATION WEAKLY OR PUTS THE DEGREE P EXTENSION ON FINITE LEVEL

THE STACKS PROJECT AUTHORS

Editorial note. All delivered mathematical text is the author's own, copied from the pinned Stacks Project chapter file more-algebra.tex; no statement or proof step was rewritten, reordered or removed. The counted statement is the labeled lemma environment at source lines 35087-35100 (canonical tag 09F4) and the counted proof is the proof environment at source lines 35102-35140, which immediately follows it and is its own proof.

Editorial note. The proof refers four times, three times to earlier lemmas of this chapter and once to Kummer theory in another chapter; all four are navigation stubs with their roles recorded in the reasons.

Editorial note. The statement and prose use the terminology of weak solutions and of extensions of finite level defined earlier in the same section; those definitions lie outside the counted spans, and the disjunctive conclusion is stated in that language exactly as the author wrote it.

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Allowed standard prerequisites (editorial).

- Kummer theory: a degree p Galois extension of a field containing a primitive p-th root of unity is obtained by adjoining the p-th root of an element of the base field. invoked at proof lines 35103-35105. Produces the element b whose p-th root generates M over L, which is the input to every case distinction in the proof.
- Weakly unramified extensions go up along a totally ramified extension, so the integral closure in the compositum remains weakly unramified over the new base. invoked at proof lines 35113-35115. Shows that the compositum of the chosen totally ramified extension with the given towers again has the weak unramifiedness required of the conclusion.
- The pre-purely-inseparable case: if the residue class of the unit is not a p-th power then the extension of discrete valuation rings is weakly unramified.

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invoked at proof lines 35128-35130. Disposes of the case in which the residue class of the unit is not a p-th power, which is one of the two alternatives in the conclusion.

Source reference. Original lemma 09DX: sources/stacks-chow/tags.txt, source lines 11880–11880. The reference is Kummer theory, invoked at proof lines 35103-35105 to write the top extension as adjoining a p-th root of an element of L; the whole case analysis built on that element is inside the counted span.

Source reference. Original lemma 09EW: sources/stacks-more-algebra/pinned-original-source.tex, source lines 34641–34641. The reference supplies the existence of the degree p separable totally ramified extension used at proof lines 35108-35111; the case analysis that consumes it is inside the counted span.

Source reference. Original lemma 09ER: sources/stacks-more-algebra/pinned-original-source.tex, source lines 34490–34490. The reference is applied at proof lines 35113-35115 to see that the integral closure in the compositum is weakly unramified over the new base; the resulting alternative of the conclusion is produced inside the counted span.

Source reference. Original lemma 09EX: sources/stacks-more-algebra/pinned-original-source.tex, source lines 34663–34663. The reference is applied at proof lines 35128-35130 to conclude weak unramifiedness when the residue class of the unit is not a p-th power; the complementary case, which yields finite level, is handled inside the counted span.

Lemma 0.2. *Let $A \subset B \subset C$ be extensions of discrete valuation rings with fractions fields $K \subset L \subset M$. Assume that*

1. *A has mixed characteristic $(0, p)$,*
2. *$A \subset B$ is weakly unramified,*
3. *B contains a primitive pth root of 1, and*
4. *M/L is Galois of degree p.*

Then there exists a finite Galois extension K_1/K totally ramified with respect to A which is either a weak solution for $A \to C$ or is such that M_1/L_1 is a degree p extension of finite level.

Proof. Let $\pi \in A$ be a uniformizer. By Kummer theory (Fields, Lemma 09DX) M is obtained from L by adjoining the root of $y^p = b$ for some $b \in L$.

If $\mathrm{ord}_B(b)$ is prime to p, then we choose a degree p separable extension K_1/K totally ramified with respect to A (for example using Lemma 09EW). Let A_1 be the integral closure of A in K_1. By Lemma 09ER the integral closure B_1 of B in $L_1 = L \otimes_K K_1$ is a discrete valuation ring weakly unramified over A_1. If K_1/K is not a weak solution for $A \to C$, then the integral closure C_1 of C in $M_1 = M \otimes_K K_1$ is a discrete valuation ring and $B_1 \to C_1$ has ramification index p. In this case, the field M_1 is obtained from L_1 by adjoining the pth root of b with $\mathrm{ord}_{B_1}(b)$ divisible by p. Replacing A by A_1, etc we may assume that $b = \pi^n u$ where $u \in B$ is a unit and n is divisible by p. Of course, in this case the extension M is obtained from L by adjoining the pth root of a unit.

Suppose M is obtained from L by adjoining the root of $y^p = u$ for some unit u of B. If the residue class of u in κ_B is not a pth power, then $B \subset C$ is weakly unramified (Lemma 09EX) and we are done. Otherwise, we can replace our choice of y by y/v where v^p and u have the same image in κ_B. After such a replacement we have

$$y^p = 1 + \pi b$$

for some $b \in B$. Then we see that $P(z) = \pi b/w^p$ where $z = (y-1)/w$. Thus we see that the extension is a degree p extension of finite level with $\xi = \pi b/w^p$. $\square$